

Medvale Internal Medicine

Chapter 2C - Pneumonia and Pulmonary Infections

Community-acquired pneumonia, hospital-acquired pneumonia, aspiration syndromes, lung abscess, empyema, and opportunistic infection

Originality and use note

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Learning goals. By the end of this segment, a learner should be able to separate pneumonia from bronchitis, pulmonary edema, pulmonary embolism, and malignancy; choose an initial site of care; identify common pathogens from exposure and host risk; start empiric therapy while obtaining the right cultures; and recognize complications that need drainage or airway support rather than another oral antibiotic.

1. Why pneumonia matters clinically

Pneumonia is an infection of the distal airspaces that becomes clinically important when inflammation fills alveoli with neutrophils, organisms, protein-rich exudate, and cellular debris. The same inflammatory process that helps contain infection also thickens the alveolar-capillary interface, creates shunt physiology, lowers oxygen saturation, and increases the work of breathing. This is why a patient with pneumonia may look stable during a clinic visit but deteriorate rapidly when respiratory reserve is limited by COPD, heart failure, immunosuppression, pregnancy, frailty, chronic kidney disease, or neurologic aspiration risk.

For internal medicine, pneumonia is both a diagnostic problem and a management problem. The diagnosis is often probabilistic: cough, fever, pleuritic pain, sputum, focal crackles, leukocytosis, hypoxemia, and a new infiltrate support the syndrome, but none is individually perfect. The management problem is time-sensitive: antibiotics should be early enough to treat bacterial disease, narrow enough to avoid preventable harm, and paired with severity assessment, oxygen support, source control, and reassessment. A graph-RAG system should therefore connect pneumonia to several relationship classes: pathogen-exposure, host-risk, radiographic-pattern, severity-triage, antibiotic-coverage, and complication-source-control.

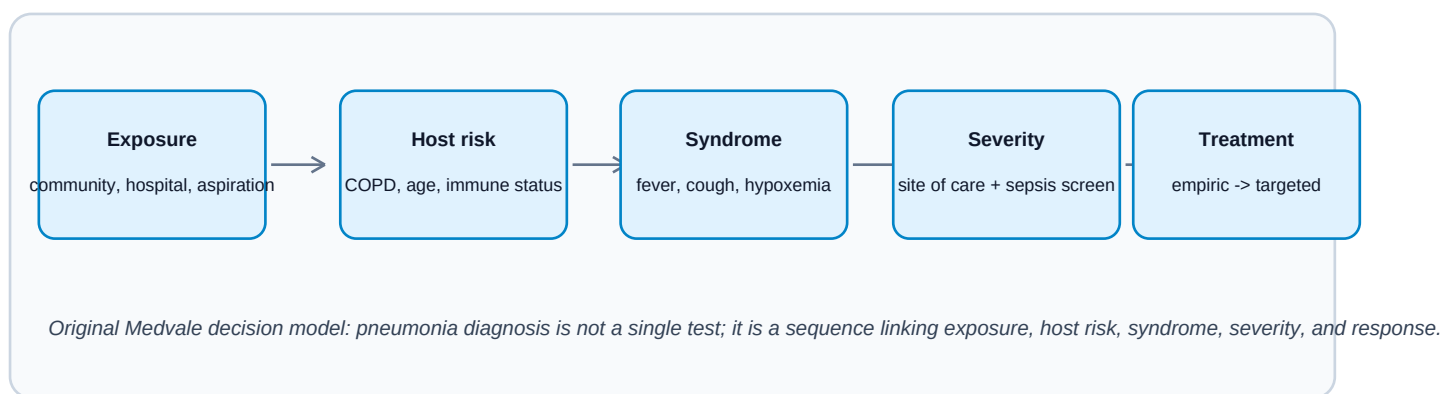


Figure 2C-1. Original Medvale framework for approaching pneumonia. The sequence moves from exposure and host risk to syndrome recognition, severity triage, and targeted treatment.

2. Core illness scripts

Syndrome	Typical patient pattern	High-yield mechanism	Numbers and anchors
Community-acquired pneumonia (CAP)	New cough, fever, dyspnea, pleuritic chest pain, sputum, and infiltrate outside the hospital or early after arrival.	Microaspiration or inhalation allows organisms to seed alveoli; neutrophilic exudate causes consolidation and shunt.	Severe CAP if 1 major criterion or at least 3 minor criteria; typical treatment duration often about 5 days if clinically stable.
Atypical pneumonia	Subacute fever, dry cough, malaise, headache, myalgias, patchy interstitial infiltrates, less dramatic auscultation.	Organisms with intracellular or nonlobar behavior provoke diffuse inflammatory injury rather than dense lobar filling.	Mycoplasma can be associated with IgM cold agglutinins; Legionella can cause hyponatremia, diarrhea, confusion, and severe disease.
Hospital-acquired pneumonia (HAP)	Pneumonia beginning at least 48 hours after hospital admission and not incubating at admission.	Hospital flora and impaired host defenses increase risk of gram-negative rods and Staphylococcus aureus.	Usual duration is 7 days when improving; local antibiogram matters.
Ventilator-associated pneumonia (VAP)	New infiltrate plus purulent secretions, fever/leukocytosis, and worsening oxygenation after at least 48 hours of mechanical ventilation.	Endotracheal tube bypasses airway defenses and permits biofilm and microaspiration.	Usual duration is 7 days when improving; obtain respiratory cultures to de-escalate.

Syndrome	Typical patient pattern	High-yield mechanism	Numbers and anchors
Aspiration pneumonitis	Abrupt respiratory symptoms after a witnessed aspiration event, often chemical injury from gastric contents.	Sterile acidic gastric contents injure alveoli and trigger inflammation without initial bacterial infection.	May improve within 24-48 hours with supportive care; antibiotics depend on clinical course.
Aspiration pneumonia	Dependent infiltrate after impaired swallowing or consciousness; fever, leukocytosis, purulent sputum, or delayed worsening.	Oropharyngeal flora infect dependent lung segments after aspiration.	Routine extra anaerobic coverage is not needed unless lung abscess or empyema is suspected.
Lung abscess	Indolent fever, night sweats, weight loss, cough with foul sputum, cavitary lesion with air-fluid level.	Necrotizing infection forms a walled cavity, often after aspiration or obstruction.	Therapy often requires prolonged antibiotics; evaluate for obstruction or malignancy if nonresolving.
Empyema	Persistent fever or sepsis with pleural fluid infection after pneumonia.	Bacteria enter pleural space; fibrin loculations prevent antibiotic penetration and lung re-expansion.	Pleural pH < 7.2, glucose < 60 mg/dL, positive Gram stain/culture, or frank pus supports drainage.

3. Pathophysiology: from colonization to respiratory failure

Most bacterial pneumonia begins when organisms reach the lower respiratory tract by microaspiration. Healthy airways usually clear aspirated organisms by cough, mucociliary transport, alveolar macrophages, secretory IgA, complement, and coordinated neutrophil recruitment. Pneumonia occurs when inoculum size, pathogen virulence, or impaired host defense overwhelms these systems. Important host-defense failures include depressed consciousness, stroke, seizures, dementia, alcohol use disorder, endotracheal intubation, neutropenia, impaired splenic function, smoking-related ciliary damage, bronchiectasis, viral epithelial injury, and structural obstruction from tumor or foreign body.

Alveolar filling produces hypoxemia mainly by low ventilation-perfusion matching and shunt. In a shunt unit, perfusion continues through alveoli that are poorly ventilated because they are filled with exudate. Supplemental oxygen helps when low V/Q units predominate, but severe shunt may respond incompletely. Tachypnea is an early compensatory response because the patient tries to maintain minute ventilation and oxygen delivery. Hypercapnia is usually late in isolated pneumonia; it suggests ventilatory failure, severe fatigue, COPD overlap, obesity hypoventilation, neuromuscular weakness, or impending need for ventilatory support.

The inflammatory response also explains extrapulmonary clues. Sepsis can cause hypotension, encephalopathy, acute kidney injury, thrombocytopenia, and lactic acidosis. Legionella may produce prominent gastrointestinal symptoms, neurologic changes, relative bradycardia, transaminitis, or hyponatremia. Mycoplasma can produce cold agglutinin hemolysis, rash, or neurologic complications. Staphylococcus aureus after influenza can produce necrotizing pneumonia with cavitation, hemoptysis, leukopenia, and rapid shock. Pneumococcus can produce lobar consolidation, bacteremia, meningitis, or empyema, especially in asplenia, alcoholism, chronic lung disease, or older age.

4. Presentation patterns and mimics

A classic bacterial pneumonia script is acute fever, rigors, productive cough, pleuritic chest pain, tachypnea, focal crackles or bronchial breath sounds, and a new lobar or segmental infiltrate. However, older adults may present with confusion, falls, anorexia, weakness, or decompensated heart failure rather than fever. Immunocompromised patients may have minimal leukocytosis or muted fever despite severe disease. A normal lung examination does not exclude pneumonia, especially early disease, dehydration, obesity, or deep lesions.

The most important mimics are pulmonary edema, pulmonary embolism, COPD/asthma exacerbation, viral bronchitis, atelectasis, lung cancer with post-obstructive change, diffuse alveolar hemorrhage, organizing pneumonia, and aspiration pneumonitis. The clinician should ask whether the process is infectious, vascular, cardiac, malignant, inflammatory, or mechanical. A patient with pleuritic pain and hemoptysis can have pneumonia, pulmonary embolism, or malignancy. A patient with bilateral opacities can have viral pneumonia, cardiogenic edema, ARDS, Pneumocystis, diffuse alveolar hemorrhage, or aspiration.

Finding	Favors pneumonia	Favors mimic
Fever and rigors	Bacterial infection or severe viral infection	Can be absent in PE, edema, malignancy; low-grade fever can occur with PE or cancer
Sputum	Purulent sputum supports bacterial airway/alveolar infection	Pink frothy sputum suggests pulmonary edema; foul sputum suggests abscess/aspiration; blood suggests PE, TB, cancer, necrotizing infection
Chest radiograph	New focal infiltrate or air bronchograms	Diffuse edema pattern, mass, wedge infarct, atelectasis volume loss, or chronic unchanged opacity
BNP/JVP/edema	May coexist but not primary pneumonia evidence	Volume overload supports heart failure exacerbation
Abrupt aspiration event	Delayed fever or persistent infiltrate favors aspiration pneumonia	Immediate hypoxemia after vomiting favors chemical pneumonitis
D-dimer/CTA context	Not a routine pneumonia test	PE workup if pleuritic pain, tachycardia, risk factors, unexplained hypoxemia

5. Diagnostic approach

The initial diagnostic goal is to confirm a compatible syndrome and determine severity. In a stable outpatient, history, vital signs, lung examination, pulse oximetry, and chest imaging are usually more useful than broad laboratory testing. In an ED or inpatient setting, labs are used to detect organ dysfunction and guide triage: complete blood count, metabolic panel, renal function, liver tests when relevant, lactate if sepsis is possible, and arterial or venous blood gas when ventilation or acid-base status is uncertain. Pulse oximetry below 90%, respiratory rate at least 30/min, systolic blood pressure below 90 mm Hg, altered mental status, multilobar disease, or rising oxygen needs should immediately change the urgency of care.

Chest radiography is the usual first imaging test because it is fast, available, and can identify focal infiltrates, multilobar disease, pleural effusion, cavitation, pneumothorax, pulmonary edema, or a mass-like opacity. CT chest is not required for every pneumonia, but it becomes valuable when the x-ray is equivocal, complications are suspected, the patient is immunocompromised, there is concern for malignancy or abscess, or the clinical severity is disproportionate to the radiograph. Lung ultrasound can identify pleural effusion, dynamic air bronchograms, consolidation near the pleura, and guide thoracentesis, but it does not replace full clinical reasoning.

Test	Best use	Do not overinterpret
Chest x-ray	Confirm infiltrate, identify multilobar disease, effusion, cavitation, edema, or pneumothorax.	A lagging infiltrate can persist after clinical improvement; early dehydration or neutropenia can make x-ray subtle.
CT chest	Evaluate complications, occult pneumonia, abscess, obstruction, immunocompromised host, or uncertain diagnosis.	Incidental nodules or chronic scarring can distract from acute physiology.
Blood cultures	Severe CAP, ICU admission, suspected MRSA/Pseudomonas, prior infection with these organisms, or recent hospitalization with IV antibiotics.	Low yield in uncomplicated outpatient CAP.
Sputum Gram stain/culture	Severe CAP, MRSA/Pseudomonas risk, nonresponse, HAP/VAP, cavitation, or immunocompromised host.	Poor-quality specimens dominated by epithelial cells reflect saliva.
Respiratory viral PCR	Influenza/COVID/RSV season, outbreak settings, immunocompromised patients, or severe pneumonia.	Positive viral test does not exclude bacterial coinfection in a toxic patient.
Urinary pneumococcal antigen	Severe CAP or when sputum is unavailable and pneumococcus is suspected.	Can remain positive after prior infection; not always needed in mild disease.
Urinary Legionella antigen	Severe CAP, outbreak/travel, hyponatremia with pneumonia, or epidemiologic risk.	Primarily detects <i>L. pneumophila</i> serogroup 1; culture/PCR may still matter.

Test	Best use	Do not overinterpret
Procalcitonin	May support stewardship trends in selected settings.	Should not be used alone to decide initial antibiotics when clinical CAP is suspected.
Thoracentesis	New significant effusion, suspected empyema, loculation, or persistent sepsis.	Small uncomplicated effusions may not be safely tappable.

6. Severity and site-of-care decision

Site of care is a safety decision. Outpatient treatment is reasonable when oxygenation, blood pressure, mentation, oral intake, comorbidities, and social support are adequate. Hospitalization is favored when the patient has hypoxemia, unstable vital signs, inability to take oral therapy, decompensated comorbidity, unreliable follow-up, severe frailty, or concern for sepsis. ICU-level care is suggested by respiratory failure requiring mechanical ventilation or septic shock requiring vasopressors. A patient can also require ICU or step-down monitoring because of rapidly rising oxygen needs, severe work of breathing, multilobar infiltrates, marked acidosis, or high-risk host factors.

Severe CAP numeric anchor

Major criteria: invasive mechanical ventilation or septic shock requiring vasopressors. Minor criteria include respiratory rate $\geq 30/\text{min}$, $\text{PaO}_2/\text{FiO}_2 \leq 250$, multilobar infiltrates, confusion, $\text{BUN} \geq 20 \text{ mg/dL}$, $\text{WBC} < 4,000/\text{mm}^3$, platelets $< 100,000/\text{mm}^3$, temperature $< 36^\circ\text{C}$, and hypotension requiring aggressive fluids. One major criterion or at least three minor criteria supports severe CAP-level management.

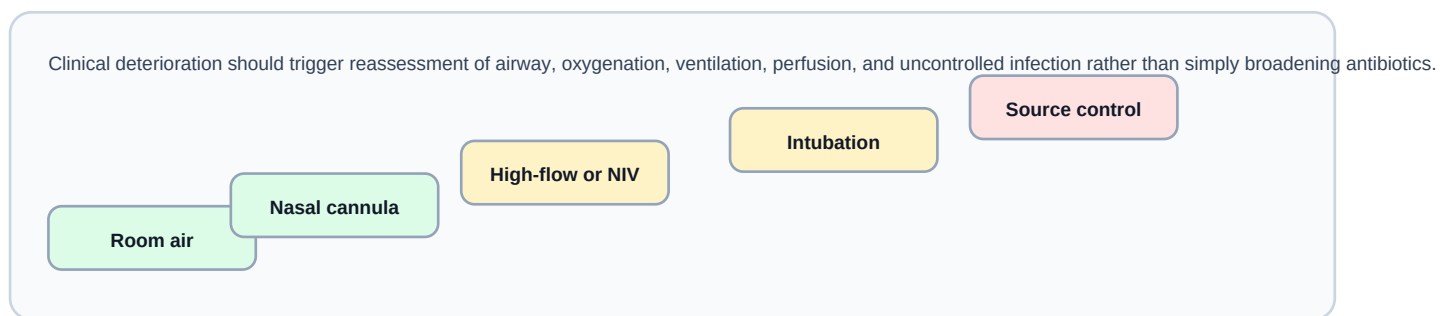


Figure 2C-2. Escalation ladder. Worsening pneumonia is not managed by antibiotics alone; airway, oxygenation, ventilation, perfusion, and source control must be reassessed.

7. Pathogen associations

Pathogen prediction should be probabilistic rather than memorized as isolated facts. *Streptococcus pneumoniae* remains a central cause of typical CAP and is classically associated with lobar consolidation, abrupt fever, pleuritic pain, rusty sputum, bacteremia, meningitis, and empyema. *Haemophilus influenzae* and *Moraxella catarrhalis* become more likely in COPD and chronic bronchitis. *Mycoplasma pneumoniae* and *Chlamydophila pneumoniae* produce atypical syndromes with dry cough and diffuse or patchy infiltrates. *Legionella* risk increases with contaminated water exposure, hotels, cruise ships, outbreaks, smoking, older age, and immunosuppression. Viral infections such as influenza, SARS-CoV-2, RSV, and human metapneumovirus can cause primary viral pneumonia and can also injure epithelium enough to permit bacterial superinfection.

Clue or host factor	Organisms to think about	Reasoning link
COPD or chronic bronchitis	<i>H. influenzae</i> , <i>Moraxella</i> , <i>S. pneumoniae</i>	Damaged mucociliary clearance and colonized airways.
Post-influenza severe pneumonia	<i>S. aureus</i> including MRSA, <i>S. pneumoniae</i>	Viral epithelial injury creates bacterial invasion risk; necrotizing disease can occur.
Bronchiectasis or cystic fibrosis	<i>Pseudomonas aeruginosa</i> , <i>S. aureus</i> , nontuberculous mycobacteria	Distorted airways retain secretions and select for resistant colonizers.

Clue or host factor	Organisms to think about	Reasoning link
Aspiration risk	Oral flora, gram-negative rods; anaerobes if abscess/empyema	Dependent inoculation follows impaired swallowing or consciousness.
Hotel/cruise/outbreak/hyponatremia/diarrhea	Legionella pneumophila	Water-associated aerosol exposure and systemic inflammatory features.
Young adult with dry cough and cold agglutinins	Mycoplasma pneumoniae	Atypical organism plus immune-mediated extrapulmonary findings.
HIV with CD4 <200 or chronic steroids	Pneumocystis jirovecii	Impaired cellular immunity; diffuse interstitial process with exertional hypoxemia.
Neutropenia or transplant	Gram-negative rods, Aspergillus, viral pathogens, Pneumocystis depending on prophylaxis	Defect in innate or cellular immunity changes the organism list.
Alcohol use disorder, poor dentition, seizure	Aspiration pathogens, Klebsiella in selected severe necrotizing cases	Reduced airway protection and large inoculum.

8. Initial management of community-acquired pneumonia

Management begins with stabilization. Give oxygen for hypoxemia, assess work of breathing, treat bronchospasm if present, obtain IV access if systemic illness is present, and screen for sepsis. Antibiotics should be started after appropriate cultures when cultures are indicated, but cultures should not delay treatment in severe illness. Reassess within 24 to 48 hours for fever curve, oxygen requirement, blood pressure, mentation, renal function, and ability to tolerate oral medication. Clinical stability usually means improving vital signs, oxygenation, mentation, oral intake, and no uncontrolled source.

Empiric antibiotic selection depends on setting, comorbidities, local resistance, allergies, and risk for MRSA or Pseudomonas. For graph-RAG, the key concept is coverage logic rather than a single memorized drug list: typical CAP coverage targets pneumococcus and common respiratory bacteria; atypical coverage targets intracellular organisms such as Mycoplasma, Chlamydia, and Legionella; MRSA coverage is added only when risk is credible; antipseudomonal coverage is added for structural lung disease, prior isolation, or recent healthcare antibiotic exposure. De-escalation occurs when cultures, PCR, clinical response, and imaging clarify the pathogen.

Scenario	Initial coverage concept	Numeric or decision anchor
Healthy outpatient without major comorbidities	Narrow CAP regimen targeting pneumococcus and common atypical pathogens depending on local resistance and patient factors.	Reassess if no improvement after 48-72 hours or if oxygenation worsens.
Outpatient with comorbidities	Broader outpatient CAP regimen that covers pneumococcus, H. influenzae, and atypicals.	Comorbidities include chronic heart, lung, liver, or renal disease; diabetes; alcoholism; malignancy; or asplenia.
Nonsevere inpatient CAP	Beta-lactam plus macrolide, or respiratory fluoroquinolone when appropriate.	Blood/sputum cultures not routine unless severe disease or resistant pathogen risk.
Severe inpatient CAP	Beta-lactam plus macrolide or beta-lactam plus respiratory fluoroquinolone; add MRSA or Pseudomonas coverage if risk factors.	Severe CAP: 1 major or ≥3 minor criteria.
Suspected influenza pneumonia	Antiviral therapy plus bacterial coinfection assessment when clinically indicated.	Do not dismiss bacterial superinfection if shock, focal lobar disease, necrosis, or rebound after viral illness.
Clinical stability achieved	Switch IV to oral when stable and absorbing medications; stop when adequate duration and stability achieved.	Typical CAP duration is often ≥5 days and guided by stability.

9. Hospital-acquired and ventilator-associated pneumonia

Hospital-acquired pneumonia is defined by timing: pneumonia developing 48 hours or more after admission and not incubating at the time of admission. Ventilator-associated pneumonia develops more than 48 hours after endotracheal intubation. Both syndromes are shaped by hospital microbial ecology, prior antibiotics, devices, aspiration, severity of illness, and local resistance patterns. Compared with routine CAP, HAP and VAP are more likely to require empiric coverage for gram-negative rods, *Pseudomonas*, and *Staphylococcus aureus* while cultures are pending.

The workup should obtain respiratory cultures whenever possible because de-escalation is a central goal. In VAP, endotracheal aspirates are often used; bronchoscopy or bronchoalveolar lavage is selected in specific clinical contexts. Clinical criteria remain important because fever, leukocytosis, infiltrate, and purulent secretions can overlap with atelectasis, pulmonary edema, ARDS, pulmonary embolism, aspiration pneumonitis, and drug fever. Once a pathogen and susceptibility profile are available, therapy should narrow. A 7-day course is a common target for HAP/VAP when the patient improves, although longer or shorter courses may be justified by source control, complications, immune status, or slow response.

Question	HAP/VAP reasoning answer
When should MRSA coverage be considered?	Prior MRSA isolation, high local MRSA prevalence, recent IV antibiotics, severe illness, or unit-specific risk.
When should double antipseudomonal coverage be considered?	High mortality risk, septic shock, ventilatory support, structural lung disease, prior IV antibiotics, or high resistance prevalence.
Why obtain cultures?	To narrow therapy, detect resistant organisms, avoid unnecessary MRSA/ <i>Pseudomonas</i> coverage, and identify noninfectious mimics.
What if cultures are negative and the patient improves?	Reassess diagnosis and consider de-escalation or stopping depending on pretest probability, specimen quality, and clinical trajectory.
What is the duration anchor?	Seven days for improving HAP/VAP is a common guideline-based anchor; adapt to clinical response and complications.

10. Aspiration syndromes and lung abscess

Aspiration is not one disease. Aspiration pneumonitis is chemical injury from gastric contents and often presents abruptly after vomiting, seizure, overdose, anesthesia, or depressed consciousness. It may cause fever, leukocytosis, and infiltrates even without bacterial infection because inflammation itself can mimic infection. The key clue is immediate respiratory compromise after a witnessed event with potential improvement over 24 to 48 hours. Supportive care is central. Antibiotics are considered when bacterial infection is likely, when symptoms persist or worsen, or when the aspiration event was not purely chemical.

Aspiration pneumonia is infection after aspirated oropharyngeal material reaches dependent lung segments. In an upright patient, the basal segments of lower lobes are common; in a supine patient, the posterior upper lobes and superior lower lobes become dependent. Risk factors include stroke, dementia, dysphagia, poor dentition, alcohol use disorder, sedatives, seizures, esophageal disease, tube feeds, and impaired cough. Routine additional anaerobic coverage is not needed for most aspiration pneumonia, but anaerobic coverage becomes important when there is lung abscess, empyema, necrotizing infection, or severe periodontal disease.

A lung abscess is a necrotic cavity within the lung parenchyma. It often has a subacute course with fever, weight loss, night sweats, malaise, and foul-smelling sputum. Radiography or CT may show a cavity with an air-fluid level. Treatment generally requires prolonged antibiotics and reassessment for obstruction, malignancy, foreign body, immunosuppression, or unusual organisms if the lesion fails to improve. Drainage is not routine for all lung abscesses because many communicate with airways and drain internally, but procedural consultation is needed for massive hemoptysis, failure of medical therapy, large abscesses, or suspected alternative diagnosis.

11. Pleural complications: parapneumonic effusion and empyema

Pneumonia can inflame the pleura and create a parapneumonic effusion. An uncomplicated parapneumonic effusion is sterile and may resolve with antibiotics alone. A complicated parapneumonic effusion has biochemical or microbiologic evidence that the pleural space is becoming infected or loculated. Empyema means pus in the pleural space. The practical management difference is that antibiotics penetrate poorly into organized pleural infection; drainage becomes therapy, not just diagnosis.

Thoracentesis should be considered when a parapneumonic effusion is more than minimal, loculated, associated with persistent fever or sepsis, or large enough to cause dyspnea. Pleural fluid tests usually include cell count with differential, protein, LDH, glucose, pH, Gram stain, bacterial culture, and sometimes cytology, triglycerides, amylase, adenosine deaminase, or fungal/mycobacterial studies depending on context. A pleural pH below 7.2, glucose below 60 mg/dL, positive Gram stain or culture, frank pus, or loculation strongly supports chest tube drainage. This chapter aligns with the broad pleural complication concepts in the reference pages that emphasize Light criteria, low pH, low glucose, empyema, and the need for drainage rather than simple observation.

Pleural condition	Pleural fluid pattern	Management logic
Uncomplicated parapneumonic effusion	Neutrophilic exudate; pH usually ≥ 7.2 ; glucose not very low; Gram stain/culture negative.	Antibiotics for pneumonia; observe effusion if small and patient improves.
Complicated parapneumonic effusion	pH < 7.2 or glucose < 60 mg/dL, high LDH, loculation, or positive Gram stain/culture.	Antibiotics plus drainage, usually chest tube.
Empyema	Frank pus or infected pleural fluid.	Drainage is mandatory; consider fibrinolytics or surgery if loculated or persistent.
Malignant effusion mimic	Often lymphocytic exudate; cytology may be positive; recurrent large effusion.	Consider cancer when nonresolving, bloody, recurrent, or with mass/weight loss.

12. Opportunistic pulmonary infections

Immunocompromised pneumonia is approached by identifying the immune defect. Neutropenia increases risk of gram-negative bacterial sepsis and invasive molds such as *Aspergillus*. Impaired cellular immunity, as in advanced HIV, transplant immunosuppression, high-dose glucocorticoids, or certain biologics, increases risk of *Pneumocystis jirovecii*, viral pneumonias, tuberculosis, nontuberculous mycobacteria, *Nocardia*, endemic fungi, and invasive molds. Humoral immune defects or asplenia increase risk of encapsulated bacteria such as *S. pneumoniae* and *H. influenzae*. Structural lung disease increases risk of colonization and infection with *Pseudomonas*, nontuberculous mycobacteria, and *Aspergillus*.

Pneumocystis pneumonia classically presents with progressive exertional dyspnea, dry cough, fever, diffuse bilateral interstitial or ground-glass opacities, and hypoxemia that may be worse than the chest x-ray suggests. HIV risk increases sharply when CD4 count is below 200 cells/mm³, but non-HIV immunosuppressed patients can present more abruptly and severely. Elevated LDH is nonspecific. Diagnosis may involve induced sputum or bronchoalveolar lavage with organism detection. Treatment is typically trimethoprim-sulfamethoxazole, with adjunctive corticosteroids when oxygenation is sufficiently impaired, such as PaO₂ below 70 mm Hg or an alveolar-arterial gradient at least 35 mm Hg in classic HIV-associated protocols.

Immune defect	Pulmonary infection pattern	Clinical implication
CD4 < 200 or prolonged steroids	<i>Pneumocystis jirovecii</i> pneumonia	Think diffuse ground-glass disease, exertional hypoxemia, TMP-SMX therapy.
Neutropenia	<i>Pseudomonas</i> , Enterobacterales, <i>Aspergillus</i>	Early broad antibacterial therapy; CT for nodules/halo; antifungal if persistent fever or imaging suggests mold.
Transplant or T-cell suppression	CMV, <i>Pneumocystis</i> , <i>Nocardia</i> , invasive fungi	Prophylaxis history and timing after transplant guide differential.
Asplenia or humoral defect	Encapsulated bacteria	Vaccination and rapid treatment of sepsis are central.

Immune defect	Pulmonary infection pattern	Clinical implication
Cavitary upper-lobe disease	TB, nontuberculous mycobacteria, endemic fungi, malignancy	Airborne isolation if TB plausible; do not assume routine CAP.

13. Prevention and health maintenance

Prevention is part of pneumonia management because many internal medicine patients have recurrent risk. Smoking cessation improves mucociliary clearance and reduces COPD exacerbations and pneumonia risk. Influenza vaccination reduces influenza pneumonia and bacterial superinfection risk. Pneumococcal vaccination recommendations have changed over time; the current adult framework is to vaccinate all adults 50 years or older with an appropriate pneumococcal conjugate vaccine if they have never received one or if status is unknown, and to vaccinate adults 19 through 49 with qualifying risk conditions. When PCV15 is used in a PCV-naïve adult schedule, PPSV23 is generally given afterward; PCV20 or PCV21 generally completes the conjugate-based adult series without PPSV23. Local policy and patient history should always be checked because prior PCV13, PCV15, PCV20, PCV21, and PPSV23 timing changes the schedule.

Aspiration prevention is equally important: evaluate dysphagia after stroke, adjust sedating medications, elevate the head of bed when appropriate, manage dental disease, avoid unnecessary acid suppression when possible, and reassess feeding tubes realistically because tube feeding does not eliminate aspiration risk. For ventilated patients, prevention bundles include minimizing sedation, daily readiness assessment, head-of-bed elevation, oral care, subglottic secretion strategies when available, and early mobilization when feasible.

14. Complications and nonresponse

Nonresponse should trigger a structured re-evaluation. The patient may have the wrong diagnosis, wrong drug, wrong dose, wrong route, resistant organism, obstructing lesion, abscess, empyema, immunocompromise, pulmonary embolism, heart failure, drug fever, or inadequate source control. Fever alone in the first 24 to 48 hours does not prove treatment failure, but worsening oxygenation, shock, rising lactate, new confusion, worsening renal function, or enlarging effusion requires urgent reassessment. Repeat imaging is appropriate if clinical response is poor or complications are suspected. Follow-up imaging after pneumonia is often considered in older smokers or patients with persistent symptoms to ensure a malignancy is not hiding behind a post-obstructive infiltrate.

Complication	Clue	Action
Acute hypoxemic respiratory failure	Rising oxygen need, PaO ₂ /FiO ₂ decline, tachypnea, fatigue	Escalate oxygen, consider high-flow/NIV/intubation depending on work of breathing and airway.
Sepsis or septic shock	Hypotension, lactate elevation, altered mentation, AKI	Cultures when indicated, early antibiotics, fluids/vasopressors as needed, source control.
Empyema	Persistent fever, pleuritic pain, effusion, low pleural pH/glucose or pus	Chest tube drainage; consider fibrinolytics or surgery if loculated.
Lung abscess	Foul sputum, cavity with air-fluid level, aspiration risk	Prolonged antibiotics; evaluate for obstruction if nonresolving.
Necrotizing pneumonia	Cavitation, hemoptysis, shock, post-influenza pattern	Consider <i>S. aureus</i> /MRSA and surgical/critical care consultation if severe.
Post-obstructive pneumonia	Recurrent same-lobe infiltrate, weight loss, hemoptysis, smoker	CT chest and bronchoscopy/biopsy pathway if suspicious.

15. High-yield exam clues

- Pneumonia plus diarrhea, confusion, hyponatremia, transaminitis, or outbreak water exposure should raise *Legionella*.
- Pneumonia plus IgM cold agglutinins, young adult, dry cough, and diffuse interstitial infiltrate should raise *Mycoplasma*.

- Post-influenza rapid cavity pneumonia, hemoptysis, shock, or leukopenia should raise *Staphylococcus aureus* including MRSA.
- Aspiration pneumonia in a supine patient favors posterior upper lobes or superior lower lobes; upright aspiration favors lower-lobe basal segments.
- Pleural fluid pH below 7.2 in a parapneumonic effusion is a drainage clue, not just an antibiotic clue.
- Hypercapnia during asthma/COPD/pneumonia overlap is a danger sign because it may reflect fatigue or ventilatory failure.
- Persistent same-location pneumonia should prompt evaluation for obstruction, foreign body, or malignancy.
- Do not use procalcitonin alone to withhold initial antibiotics when clinical and radiographic CAP is convincing.
- A positive viral test does not rule out bacterial coinfection in severe, focal, necrotizing, or shock-associated pneumonia.
- Routine extra anaerobic coverage for aspiration pneumonia is unnecessary unless abscess, empyema, or necrotizing infection is suspected.

16. Common pitfalls

Pitfall	Why it is dangerous	Safer habit
Calling every cough pneumonia	Bronchitis, asthma/COPD exacerbation, PE, pulmonary edema, and malignancy can be mislabeled.	Require a syndrome plus imaging or strong clinical rationale.
Ignoring oxygen trend	A patient can be normotensive but heading toward respiratory failure.	Track respiratory rate, work of breathing, SpO ₂ , oxygen device, and mental status.
Treating empyema with antibiotics alone	Organized pleural pus has poor antibiotic penetration.	Use thoracentesis and chest tube drainage when fluid criteria support it.
Overusing broad-spectrum antibiotics	Increases <i>C. difficile</i> , resistance, kidney injury, and diagnostic confusion.	Add MRSA/ <i>Pseudomonas</i> coverage only with credible risk and de-escalate.
Missing aspiration pneumonitis	Chemical injury may be mistaken for bacterial infection requiring prolonged antibiotics.	Use time course: abrupt after aspiration and improvement in 24-48 hours supports pneumonitis.
Missing post-obstructive cancer	Repeated same-lobe pneumonia may be the first cancer clue.	CT and bronchoscopy pathway when recurrent, nonresolving, or suspicious.

17. Practice questions

Question. A 71-year-old man with COPD has fever, productive cough, right lower-lobe infiltrate, SpO₂ 88% on room air, and BUN 32 mg/dL. What early decision matters most?

Answer. Determine site of care and severity before focusing on the exact organism. Hypoxemia and elevated BUN suggest higher risk; oxygen support, inpatient evaluation, and empiric CAP therapy are more urgent than outpatient reassurance.

Question. A 33-year-old has dry cough, low-grade fever, diffuse patchy infiltrates, and cold agglutinin hemolysis. What organism is suggested?

Answer. *Mycoplasma pneumoniae*. The association is atypical pneumonia plus IgM cold agglutinins.

Question. A patient with pneumonia remains febrile after 4 days and has a moderate pleural effusion. Pleural fluid pH is 7.05 and glucose is 38 mg/dL. What is the key management principle?

Answer. This is a complicated parapneumonic effusion/empyema pattern. Antibiotics alone are insufficient; drainage with a chest tube is usually needed.

Question. A nursing-home patient aspirates during a seizure and develops immediate hypoxemia, fever, leukocytosis, and a new dependent opacity. The patient improves markedly within 24 hours. What diagnosis is most likely?

Answer. Aspiration pneumonitis. Chemical injury can cause fever and leukocytosis, but rapid improvement after a witnessed event supports pneumonitis rather than established bacterial aspiration pneumonia.

Question. A ventilated ICU patient develops new purulent secretions, fever, worsening oxygenation, and a new infiltrate after 5 days of intubation. What is the treatment-duration anchor if he improves?

Answer. Ventilator-associated pneumonia should be treated empirically according to local resistance and cultures, then narrowed. Seven days is a common duration anchor if clinical response is adequate.

Question. A 58-year-old smoker has recurrent pneumonia in the same right upper-lobe region and 8-pound weight loss. What should be considered?

Answer. Post-obstructive pneumonia from lung cancer or another obstructing lesion. CT chest and possible bronchoscopy are appropriate rather than repeated empiric outpatient antibiotics alone.

18. Graph-RAG concept map seeds

The following entities and relationships are intentionally written in explicit triples to support extraction into a graph database. Pneumonia causes alveolar exudate. Alveolar exudate causes shunt physiology. Shunt physiology causes hypoxemia. Severe CAP is defined by mechanical ventilation, vasopressor-dependent shock, or multiple minor severity criteria. COPD increases risk of H. influenzae pneumonia. Influenza increases risk of secondary Staphylococcus aureus pneumonia. Bronchiectasis increases risk of Pseudomonas infection. Aspiration risk causes dependent pneumonia. Lung abscess requires prolonged antibiotics and evaluation for obstruction if nonresolving. Empyema requires pleural drainage. Pleural pH below 7.2 predicts need for drainage in parapneumonic effusion. HAP and VAP require local antibiogram awareness. Culture data enables antibiotic de-escalation. Pneumococcal vaccination prevents pneumococcal pneumonia and invasive pneumococcal disease. Persistent same-lobe pneumonia suggests obstructing malignancy.

Selected source anchors used for medical accuracy

1. ATS/IDSA. Diagnosis and Treatment of Adults with Community-acquired Pneumonia. 2019 guideline and associated clinical pathway.
2. IDSA/ATS. Management of Adults With Hospital-acquired and Ventilator-associated Pneumonia. 2016 guideline.
3. CDC. Pneumococcal vaccine recommendations for adults and risk-based schedules, updated 2025-2026.
4. User-provided pulmonology reference pages were used only as coverage signals for pleural effusion, empyema, pneumothorax, lung neoplasms, solitary pulmonary nodules, and mediastinal masses; no proprietary images, tables, or wording were reproduced.